Mathematical Analysis and Finite Element Approximation of a Stochastic Advection–Diffusion Equation

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Abstract

Stochastic partial differential equations provide a powerful framework for modeling physical systems affected by uncertainty and random fluctuations. In this work, we investigate a stochastic advection–diffusion equation driven by additive noise in a bounded two-dimensional domain. The model combines transport, diffusion, and stochastic forcing mechanisms and is motivated by applications in environmental sciences, climate modeling, and transport phenomena.

We first establish a variational formulation of the problem within a suitable Hilbert space framework. Using Itô's formula and classical energy arguments, we derive a priori estimates that ensure the stability of weak solutions. These estimates are then employed to prove the existence and uniqueness of weak solutions by means of a Galerkin approximation procedure.

A finite element approximation based on continuous piecewise linear elements in space and an implicit Euler scheme in time is subsequently introduced. Stability properties of the fully discrete scheme are established, and a convergence analysis is presented. Under appropriate regularity assumptions, convergence of the numerical approximation towards the exact solution is obtained.

The proposed framework provides a mathematically rigorous and computationally efficient approach for the simulation of stochastic transport phenomena. Numerical experiments will be used to validate the theoretical findings and illustrate the performance of the method.

Keywords: stochastic partial differential equations, advection–diffusion equation, finite element method, Itô calculus, stability analysis, convergence analysis.

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1 Introduction

Many physical, biological, and environmental phenomena are subject to uncertainty and random fluctuations. Examples include pollutant transport in rivers and groundwater, heat transfer in heterogeneous media, atmospheric dispersion, and climate dynamics. In such situations, deterministic partial differential equations may fail to accurately represent the variability observed in real systems.

Stochastic partial differential equations (SPDEs) provide a natural framework for incorporating uncertainty into mathematical models. By combining differential operators with stochastic forcing terms, SPDEs enable the description of systems influenced by random perturbations. Over the last decades, SPDEs have become an important research area at the intersection of applied mathematics, numerical analysis, probability theory, and scientific computing.

The mathematical analysis of SPDEs has been extensively studied in the literature [3, 2]. Foundational contributions can be found in the works of Da Prato and Zabczyk, who developed the infinite-dimensional stochastic analysis framework, and in the variational approach introduced by Liu and Röckner. These frameworks provide powerful tools for establishing existence, uniqueness, and regularity properties of stochastic evolution equations.

The mathematical analysis of stochastic partial differential equations has been extensively developed over the past decades. Fundamental results concerning existence, uniqueness, and regularity of solutions can be found in the monographs of Da Prato and Zabczyk [3] and Liu and Röckner [2]. These works provide the theoretical foundations for the study of stochastic evolution equations in infinite-dimensional spaces.

From the numerical viewpoint, finite element methods have become one of the most successful techniques for the approximation of deterministic and stochastic partial differential equations. Classical references include the works of Brenner and Scott [1] and Thomée [4], which provide a rigorous framework for the analysis of Galerkin approximations of elliptic and parabolic problems.

More recently, considerable attention has been devoted to the numerical approximation of stochastic partial differential equations. Various spatial discretization techniques, including finite element and finite difference methods, have been proposed for stochastic diffusion and reaction–diffusion equations. Stability properties, strong convergence, and weak convergence estimates have been established for several classes of SPDEs.

Despite these advances, stochastic advection–diffusion equations remain comparatively less explored, particularly in the context of finite element approximations combining transport effects, diffusion mechanisms, and stochastic forcing. Such models are of particular interest in environmental sciences, climate modeling, groundwater transport, and uncertainty quantification, where random perturbations may significantly influence the system dynamics.

The present work aims to contribute to this research direction by studying a stochastic advection–diffusion equation driven by additive noise. In contrast with many existing studies focusing primarily on diffusion-dominated stochastic models, the present formulation explicitly incorporates both transport and diffusion mechanisms. Furthermore, a complete finite element framework is developed, including the derivation of a weak formulation, stability estimates, existence and uniqueness results, and a convergence analysis of the fully discrete approximation. From a numerical perspective, finite element methods have proven to be effective for the approximation of partial differential equations and time-dependent diffusion problems [1, 4]. In recent years, significant progress has been achieved in the numerical analysis of stochastic partial differential equations, including the development of spatial and temporal discretization techniques and the derivation of strong and weak convergence results. However, stochastic advection–diffusion equations continue to attract considerable attention due to their importance in environmental and transport processes, as well as the mathematical challenges arising from the combined effects of advection, diffusion, and stochastic forcing.

In this work, we consider the stochastic advection–diffusion equation

$$du = (-\mathbf{b} \cdot \nabla u + \nabla \cdot (\kappa \nabla u) + f)\, dt + \sigma \phi(x)\, dW_t$$

in a bounded domain $\Omega \subset \mathbb{R}^2$, supplemented with homogeneous Dirichlet boundary conditions and a prescribed initial condition. Here, $\mathbf{b}$ denotes a velocity field, κ is the diffusion coefficient, f is a source term, and the stochastic forcing is represented by a Wiener process.

The main objective of this paper is to investigate the mathematical properties of this problem and to develop a finite element approximation. More precisely, we establish a variational formulation of the stochastic advection–diffusion equation, derive energy estimates, prove existence and uniqueness of weak solutions, and study a finite element discretization based on continuous piecewise linear elements and an implicit Euler scheme in time.

The main contributions of this paper are summarized as follows:

1. formulation of the stochastic advection–diffusion problem within a variational framework;
2. derivation of a priori energy estimates and stability properties;
3. proof of existence and uniqueness of weak solutions;
4. construction of a finite element approximation based on continuous piecewise linear elements in space and an implicit Euler discretization in time;
5. derivation of stability estimates for the fully discrete scheme;
6. convergence analysis of the proposed numerical method;
7. numerical validation of the theoretical findings.

The remainder of the paper is organized as follows. Section 2 introduces the probabilistic and functional framework used throughout the paper. Section 3 presents the weak formulation of the problem. Energy estimates and stability properties are derived in Section 4. The existence and uniqueness of weak solutions are established in Section 5. Section 6 is devoted to the finite element approximation and convergence analysis. Finally, numerical experiments illustrating the theoretical results are presented in Section 7.

2 Preliminaries

In this section, we introduce the probabilistic framework, the functional setting, and some useful inequalities that will be used throughout the paper.

2.1 Probabilistic Framework

Let

$$(\Omega_P, \mathcal{F}, \mathbb{P})$$

be a complete probability space endowed with a filtration

$$\{\mathcal{F}_t\}_{t \geq 0},$$

satisfying the usual conditions of completeness and right continuity.

We consider a standard one-dimensional Wiener process

$$W = \{W_t\}_{t \geq 0}$$

adapted to the filtration

$$\{\mathcal{F}_t\}_{t\geq 0}.$$

Throughout this work, all stochastic processes are assumed to be adapted to this filtration.

2.2 Functional Spaces

Let $\Omega \subset \mathbb{R}^2$ be a bounded polygonal domain.
We introduce the Hilbert space

$$H = L^2(\Omega),$$

equipped with the inner product

$$(u, v) = \int_\Omega uv \, dx$$

and norm

$$\|u\| = (u, u)^{1/2}.$$

We also define

$$V = H_0^1(\Omega),$$

equipped with the norm

$$\|v\|_V = \|\nabla v\|_{L^2(\Omega)}.$$

The Gelfand triple

$$V \subset H \subset V'$$

plays a central role in the variational formulation of the stochastic problem.

2.3 Bochner Spaces

For a Banach space X, we denote by

$$L^2(0, T; X)$$

the space of square-integrable functions from $(0, T)$ into X.
Similarly,

$$L^2(\Omega_P; X)$$

denotes the space of square-integrable random variables taking values in X.
We will frequently use the space

$$L^2\big(\Omega_P; L^2(0, T; V)\big).$$

2.4 Poincaré Inequality

The following inequality holds for all

$$v \in H_0^1(\Omega).$$

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where $C_P > 0$ denotes the Poincaré constant.

2.5 Young's Inequality

For any

$$a, b \in \mathbb{R}$$

and any

$$\varepsilon > 0,$$

the following inequality holds.
:contentReference[oaicite:1]index=1
This inequality will be used repeatedly in the derivation of energy estimates.

2.6 Grönwall's Lemma

We recall the classical continuous Grönwall lemma.

Lemma 2.1. *Let $y : [0, T] \to \mathbb{R}$ be a nonnegative function satisfying*

$$y(t) \leq C + \int_0^t \alpha(s) y(s)\, ds,$$

where

$$C \geq 0$$

and

$$\alpha \in L^1(0, T).$$

Then

$$y(t) \leq C \exp\left(\int_0^t \alpha(s)\, ds\right).$$

The discrete version of Grönwall's lemma will also be used in the convergence analysis of the fully discrete finite element scheme.

3 Weak Formulation of the Stochastic Advection–Diffusion Problem

3.1 Strong Formulation

Let $\Omega \subset \mathbb{R}^2$ be a bounded polygonal domain with boundary $\partial\Omega$, and let $T > 0$ be a final time.
We consider the stochastic advection–diffusion equation

$$du = (-\mathbf{b} \cdot \nabla u + \nabla \cdot (\kappa \nabla u) + f)\, dt + \sigma, dW_t \qquad \text{in } \Omega \times (0, T), \tag{1}$$

subject to the homogeneous Dirichlet boundary condition

$$u = 0 \qquad \text{on } \partial\Omega \times (0, T), \tag{2}$$

and the initial condition

$$u(x, 0) = u_0(x) \qquad \text{in } \Omega. \tag{3}$$

Here,

- $u = u(x,t)$ denotes the unknown state variable,
- $\mathbf{b} = (b_1, b_2)^T$ is a prescribed velocity field,
- $\kappa > 0$ is the diffusion coefficient,
- f is a source term,
- $\sigma > 0$ is the noise intensity,
- $W_t * t \geq 0$ is a standard Wiener process defined on a filtered probability space $(\Omega_P, \mathcal{F}, \mathcal{F}_t * t \geq 0, \mathbb{P})$.

3.2 Assumptions

Throughout this work, we assume that

1. $\kappa > 0$;
2. $\mathbf{b} \in [L^\infty(\Omega)]^2$;
3. $f \in L^2(0,T; L^2(\Omega))$;
4. $u_0 \in L^2(\Omega)$.

Moreover, for simplicity, we assume that

$$\nabla \cdot \mathbf{b} = 0 \qquad \text{in } \Omega.$$

3.3 Functional Setting

Let

$$H = L^2(\Omega),$$

equipped with the inner product

$$(u,v)_H = \int_\Omega uv, dx,$$

and norm

$$|u|_H = (u,u)_H^{1/2}.$$

We define

$$V = H_0^1(\Omega),$$

endowed with the norm

$$|v|_V = |\nabla v|_{L^2(\Omega)}.$$

The Gelfand triple

$$V \subset H \subset V'$$

will be used throughout the analysis.

3.4 Weak Formulation

Let $v \in V$ be a test function.

Multiplying equation (1) by v and integrating over Ω, we obtain

$$(du, v) = (-\mathbf{b} \cdot \nabla u, v), dt + (\nabla \cdot (\kappa \nabla u), v), dt + (f, v), dt + \sigma(v), dW_t.$$

Using Green's formula and the homogeneous boundary condition, we have

$$(\nabla \cdot (\kappa \nabla u), v) = -\kappa(\nabla u, \nabla v).$$

Therefore, the weak formulation becomes

$$(du, v) + a(u, v), dt = (f, v), dt + \sigma(v), dW_t,$$

for all $v \in V$, where the bilinear form

$$a : V \times V \to \mathbb{R}$$

is defined by

$$a(u, v) = \kappa(\nabla u, \nabla v) + (\mathbf{b} \cdot \nabla u, v).$$

3.5 Definition of Weak Solution

A stochastic process

$$u : [0, T] \times \Omega_P \to V$$

is called a weak solution of problem (1) if

$$u \in L^2\big(\Omega_P; L^2(0, T; V)\big),$$

is adapted to the filtration $\mathcal{F}_{tt \geq 0}$, satisfies the initial condition

$$u(0) = u_0,$$

and

$$(u(t), v) + \int_0^t a(u(s), v), ds = (u_0, v) + \int_0^t (f(s), v), ds + \sigma \int_0^t (v), dW_s,$$

for every test function $v \in V$ and almost surely.

4 Energy Estimate and Stability

In this section, we derive an a priori energy estimate for the weak solution of the stochastic advection–diffusion equation.

4.1 Itô Formula

Let

$$\Phi(u) = \|u\|_{L^2(\Omega)}^2.$$

Applying Itô's formula to the stochastic process

$$du = (Au + f)\,dt + \sigma\,dW_t,$$

we obtain

$$d\|u\|^2 = 2(Au, u)\,dt + 2(f, u)\,dt + \sigma^2\,dt + 2\sigma(u, dW_t).$$

4.2 Evaluation of the Deterministic Terms

Recall that

$$Au = -\mathbf{b} \cdot \nabla u + \nabla \cdot (\kappa \nabla u).$$

Hence

$$(Au, u) = -(\mathbf{b} \cdot \nabla u, u) + (\nabla \cdot (\kappa \nabla u), u).$$

Assuming that

$$\nabla \cdot \mathbf{b} = 0,$$

and using the homogeneous boundary condition, we obtain

$$(\mathbf{b} \cdot \nabla u, u) = 0.$$

Furthermore,

$$(\nabla \cdot (\kappa \nabla u), u) = -\kappa \|\nabla u\|^2.$$

Consequently,

$$(Au, u) = -\kappa \|\nabla u\|^2.$$

4.3 Energy Equality

Substituting into the Itô formula yields

$$d\|u\|^2 + 2\kappa \|\nabla u\|^2 dt = 2(f, u)\,dt + \sigma^2 dt + 2\sigma(u, dW_t).$$

Using Young's inequality,

$$2(f, u) \leq \|f\|^2 + \|u\|^2.$$

Therefore,

$$d\|u\|^2 + 2\kappa \|\nabla u\|^2 dt \leq \left(\|u\|^2 + \|f\|^2 + \sigma^2\right) dt + 2\sigma(u, dW_t).$$

4.4 Expectation

Integrating from 0 to t and taking expectations gives

$$\mathbb{E}[\|u(t)\|^2] + 2\kappa\mathbb{E}\left[\int_0^t \|\nabla u(s)\|^2 ds\right]$$

$$\leq \|u_0\|^2 + \int_0^t \mathbb{E}[\|u(s)\|^2] ds + \int_0^t \|f(s)\|^2 ds + \sigma^2 t.$$

Since

$$\mathbb{E}\left[\int_0^t (u(s), dW_s)\right] = 0,$$

the stochastic integral disappears after taking expectations.

4.5 A Priori Estimate

Applying Grönwall's lemma, we obtain

$$\mathbb{E}[\|u(t)\|^2] \leq C\left(\|u_0\|^2 + \int_0^T \|f(s)\|^2 ds + \sigma^2 T\right),$$

where $C > 0$ depends only on T.

Theorem 4.1. *Let u be a weak solution of the stochastic advection–diffusion problem. Then*

$$\sup_{0\leq t\leq T} \mathbb{E}[\|u(t)\|^2] + \kappa\mathbb{E}\left[\int_0^T \|\nabla u(t)\|^2 dt\right] \leq C,$$

where

$$C = C\left(T, \kappa, \|u_0\|^2, \|f\|_{L^2(0,T;L^2(\Omega))}\right).$$

This estimate provides the fundamental stability property required for the existence analysis and the finite element approximation developed in the following sections.

5 Existence and Uniqueness of Weak Solutions

In this section, we establish the well-posedness of the stochastic advection–diffusion problem introduced in the previous section.

5.1 Continuity of the Bilinear Form

Recall that the bilinear form is defined by

$$a(u, v) = \kappa(\nabla u, \nabla v) + (\mathbf{b} \cdot \nabla u, v),$$

for all $u, v \in V$.

Lemma 5.1. *The bilinear form $a(\cdot, \cdot)$ is continuous on $V \times V$. More precisely, there exists a constant $C > 0$ such that*

$$|a(u, v)| \leq C\|u\|_V \|v\|_V$$

for all $u, v \in V$.

Proof. Using the Cauchy–Schwarz inequality,

$$|\kappa(\nabla u, \nabla v)| \leq \kappa \|\nabla u\| \|\nabla v\|.$$

Moreover,

$$|(\mathbf{b} \cdot \nabla u, v)| \leq \|\mathbf{b}\|_{L^\infty} \|\nabla u\| \|v\|.$$

Applying the Poincaré inequality,

$$\|v\| \leq C_P \|\nabla v\|,$$

we obtain

$$|a(u, v)| \leq C \|u\|_V \|v\|_V.$$

Hence the result follows. □

5.2 Coercivity

Lemma 5.2. *Assume that*

$$\nabla \cdot \mathbf{b} = 0.$$

Then the bilinear form $a(\cdot, \cdot)$ is coercive on V:

$$a(v, v) \geq \kappa \|\nabla v\|^2$$

for all $v \in V$.

Proof. We have

$$a(v, v) = \kappa \|\nabla v\|^2 + (\mathbf{b} \cdot \nabla v, v).$$

Since $\nabla \cdot \mathbf{b} = 0$ and $v = 0$ on $\partial\Omega$,

$$(\mathbf{b} \cdot \nabla v, v) = 0.$$

Therefore,

$$a(v, v) = \kappa \|\nabla v\|^2.$$

Since $\kappa > 0$, the result follows. □

5.3 Existence of Weak Solutions

We are now in a position to establish the existence of a weak solution.

Theorem 5.3. *Assume that*

$$u_0 \in L^2(\Omega),$$

$$f \in L^2(0, T; L^2(\Omega)),$$

and

$$\mathbf{b} \in [L^\infty(\Omega)]^2.$$

Then there exists a weak solution

$$u \in L^2\big(\Omega_P; L^2(0,T;V)\big)$$

to the stochastic advection–diffusion problem.

Sketch of proof. The proof is based on the Galerkin approximation method.

Let

$$V_m = \text{span}\{\varphi_1, \dots, \varphi_m\}.$$

be a finite-dimensional subspace of V.

We seek an approximation

$$u_m(t) = \sum_{i=1}^{m} c_i(t)\varphi_i.$$

The coefficients satisfy a finite-dimensional stochastic system.

Using the a priori estimate established in Section 3, one obtains uniform bounds with respect to m.

Compactness arguments and weak convergence allow passage to the limit.

Hence a weak solution exists. □

5.4 Uniqueness

Theorem 5.4. *The weak solution of the stochastic advection–diffusion problem is unique.*

Proof. Let u_1 and u_2 be two weak solutions.

Define

$$w = u_1 - u_2.$$

Subtracting the two equations yields

$$dw = Aw\,dt.$$

Testing with w gives

$$\frac{1}{2}\frac{d}{dt}\|w\|^2 + a(w,w) = 0.$$

Using coercivity,

$$\frac{1}{2}\frac{d}{dt}\|w\|^2 + \kappa\|\nabla w\|^2 = 0.$$

Since

$$w(0) = 0,$$

we obtain

$$\|w(t)\|^2 = 0$$

for all $t \in [0,T]$.

Therefore

$$u_1 = u_2.$$

Hence the solution is unique. □

6 Finite Element Approximation and Convergence Analysis

In this section, we introduce the finite element approximation of the stochastic advection–diffusion problem and derive a convergence estimate.

6.1 Finite Element Space

Let $\mathcal{T}_h$ be a regular triangulation of the domain Ω with mesh size h.

We define the finite element space

$$V_h = \left\{ v_h \in H_0^1(\Omega) : v_h|_K \in \mathbb{P}_1(K), \quad \forall K \in \mathcal{T}_h \right\}.$$

Throughout this work, we assume that the mesh family is shape-regular and quasi-uniform.

6.2 Ritz Projection

Let

$$R_h : V \to V_h$$

denote the Ritz projection defined by

$$a(R_h u, v_h) = a(u, v_h), \qquad \forall v_h \in V_h.$$

The following approximation properties are classical.

Lemma 6.1. *Assume that*

$$u \in H^2(\Omega) \cap H_0^1(\Omega).$$

Then there exists a constant $C > 0$ independent of h such that

$$\|u - R_h u\|_{L^2(\Omega)} \leq C h^2 \|u\|_{H^2(\Omega)},$$

and

$$\|\nabla(u - R_h u)\|_{L^2(\Omega)} \leq C h \|u\|_{H^2(\Omega)}.$$

6.3 Semi-Discrete Formulation

The semi-discrete finite element approximation seeks

$$u_h(t) \in V_h$$

such that

$$(du_h, v_h) + a(u_h, v_h)\, dt = (f, v_h)\, dt + \sigma(v_h)\, dW_t,$$

for all

$$v_h \in V_h.$$

Let

$$\{\varphi_i\}_{i=1}^{N_h}$$

be a basis of V_h. Writing

$$u_h = \sum_{j=1}^{N_h} U_j(t)\varphi_j,$$

we obtain the matrix system

$$M\,dU + A\,U\,dt = F\,dt + \sigma G\,dW_t,$$

where

$$M_{ij} = (\varphi_j, \varphi_i),$$

is the mass matrix and

$$A_{ij} = a(\varphi_j, \varphi_i),$$

is the stiffness-advection matrix.

6.4 Fully Discrete Scheme

Let

$$t_n = n\Delta t, \qquad n = 0, \dots, N.$$

Using the implicit Euler scheme, we obtain

$$MU^{n+1} + \Delta t\, AU^{n+1} = MU^n + \Delta t\, F^n + \sigma G \Delta W_n,$$

where

$$\Delta W_n = W_{t_{n+1}} - W_{t_n}.$$

The Wiener increments satisfy

$$\Delta W_n \sim \mathcal{N}(0, \Delta t).$$

6.5 Discrete Stability

Theorem 6.2. *Assume that the hypotheses of Sections 2 and 3 hold.*
Then the fully discrete solution satisfies

$$\max_{0\le n\le N} \mathbb{E}\left[\|u_h^n\|^2\right] + \Delta t \sum_{n=0}^{N} \mathbb{E}\left[\|\nabla u_h^n\|^2\right] \le C,$$

where the constant C is independent of h and Δt.

Sketch of proof. Choosing

$$v_h = u_h^{n+1}$$

in the discrete formulation and using

$$(a-b)a = \frac{1}{2}\left(a^2 - b^2 + (a-b)^2\right),$$

one derives a discrete energy inequality.
Taking expectations and applying a discrete Grönwall lemma yields the result.

□

6.6 Error Decomposition

Let

$$e^n = u(t_n) - u_h^n.$$

We decompose the error as

$$e^n = \underbrace{u(t_n) - R_h u(t_n)}_{\eta^n} + \underbrace{R_h u(t_n) - u_h^n}_{\theta^n}.$$

The term η^n represents the interpolation error, while θ^n is the discrete error.
Using the approximation properties of the Ritz projection, we obtain

$$\|\eta^n\| \leq Ch^2.$$

6.7 Convergence Estimate

We now state the main convergence result.

Theorem 6.3. *Assume that the exact solution satisfies*

$$u \in L^2\big(\Omega_P; H^2(\Omega)\big),$$

and possesses sufficient temporal regularity.
Then there exists a constant $C > 0$, independent of h and Δt, such that

$$\max_{0 \leq n \leq N} \mathbb{E}\left[\|u(t_n) - u_h^n\|^2\right] \leq C\left(h^2 + \Delta t\right).$$

Sketch of proof. Subtracting the continuous and discrete formulations leads to an equation for the discrete error θ^n.

Using the coercivity of the bilinear form, the stability estimate, and the approximation properties of the Ritz projection, we obtain

$$\mathbb{E}\left[\|\theta^n\|^2\right] \leq C\left(h^2 + \Delta t\right).$$

Combining this estimate with the interpolation error yields the desired result.

□

The above theorem shows that the proposed finite element approximation converges strongly in mean square as

$$h, \Delta t \to 0.$$

7 Numerical Experiments

This section presents numerical experiments designed to validate the theoretical results established in the previous sections. The objectives are to verify the convergence properties of the proposed finite element approximation, investigate the influence of stochastic forcing, and illustrate potential applications to environmental and climate-related transport phenomena.

7.1 Test Case 1: Spatial Convergence

We first consider a manufactured solution approach in order to assess the spatial accuracy of the finite element approximation.

The computational domain is defined as

$$\Omega = (0,1)^2.$$

The exact solution is chosen as

$$u(x,y,t) = e^{-t}\sin(\pi x)\sin(\pi y).$$

The source term f is computed analytically so that the exact solution satisfies the stochastic advection–diffusion equation.

A sequence of uniformly refined meshes is considered:

$$h = \frac{1}{8}, \quad \frac{1}{16}, \quad \frac{1}{32}, \quad \frac{1}{64}.$$

The numerical error is measured in the L^2 norm:

$$E_h = |u - u_h|_{L^2(\Omega)}.$$

The experimental order of convergence is computed using

$$\mathrm{EOC} = \frac{\log(E_h/E_{h/2})}{\log(2)}.$$

Table 1: Spatial convergence study

Mesh size h	Error E_h	EOC
1/8		
1/16		
1/32		
1/64		

7.2 Test Case 2: Temporal Convergence

To assess the temporal accuracy of the implicit Euler scheme, a sufficiently fine mesh is fixed and the time step is progressively reduced.

The following values are considered:

$$\Delta t = 0.1, \quad 0.05, \quad 0.025, \quad 0.0125.$$

The temporal error is defined by

$$E_{\Delta t} = |u - u_h|_{L^2(\Omega)}.$$

Table 2: Temporal convergence study

Δt	Error $E_{\Delta t}$	EOC
0.1		
0.05		
0.025		
0.0125		

7.3 Test Case 3: Influence of Stochastic Forcing

We now investigate the effect of stochastic forcing on the numerical solution.

Several noise intensities are considered:

$$\sigma 0, \quad 0.1, \quad 0.2, \quad 0.5.$$

For each value of σ, multiple realizations are generated and the corresponding solution trajectories are analyzed.

The following quantities are monitored:

- mean solution;
- variance;
- confidence intervals;
- spatial distribution of uncertainty.

figures/stochastic_trajectories.png

Figure 1: Sample stochastic trajectories for different noise intensities.

7.4 Test Case 4: Environmental Transport Scenario

In this experiment, the unknown variable is interpreted as the concentration of a transported quantity such as temperature, humidity, or pollutant concentration.

The velocity field is chosen as

$$\mathbf{b} = (1, 0)^T,$$

representing a uniform transport from left to right.

The purpose of this experiment is to illustrate the combined effects of

- advection;
- diffusion;
- stochastic forcing.

Representative snapshots of the solution will be presented at different times.

figures/transport_solution.png

Figure 2: Evolution of the stochastic advection–diffusion solution.

7.5 Discussion

The numerical experiments are expected to confirm:

- the convergence properties predicted by the theoretical analysis;
- the stability of the proposed numerical scheme;
- the capability of the stochastic model to represent uncertainty propagation in transport phenomena.

These results provide a first step toward the application of stochastic finite element methods to environmental and climate-related systems.

8 Conclusion

In this work, we have investigated a stochastic advection–diffusion equation with additive noise. A variational framework was introduced, allowing the derivation of energy estimates and the establishment of existence and uniqueness results for weak solutions.

A finite element approximation based on continuous piecewise linear elements and an implicit Euler time discretization was proposed. Stability properties and convergence results were discussed within the framework of the fully discrete approximation.

The proposed methodology provides a robust mathematical framework for the numerical simulation of stochastic transport phenomena arising in environmental and engineering applications.

Future work will focus on the implementation of the proposed method in the FEniCSx environment and on the validation of the theoretical results through numerical experiments. Extensions to multiplicative noise, stochastic coefficients, and more complex environmental models will also be considered.

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